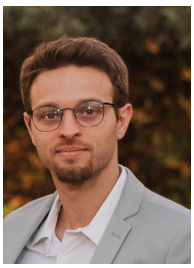
Guy Bar-Shalom

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https://barsguy.github.io/



Education

2022 – 2026 **PhD Computer Science.**
Technion - Israel Institute of Technology.
Advisors Prof. Ran El-Yaniv, and Prof. Haggai Maron.
My research focuses on Geometric Deep Learning, leveraging data symmetries to enhance model learning. I explore applications in graph neural networks and uncertainty estimation in both vision models and large language models.

2020 – 2022 **MSc Computer Science.**
Advisor Prof. Ran El-Yaniv.
Technion - Israel Institute of Technology.
Uncertainty Estimation in Vision Models.

2016 – 2020 **BSc Computer Science and Physics.**
Technion - Israel Institute of Technology.

Employment

2025 **Meta.**
Research Intern
Worked on weight space models (paper under review at ICLR 2026)

2023 **Google-Verily.**
Research Intern
Worked on video analysis (published at WACV 2024)

2018 – present **Teaching assistant (Technion).**
Algorithms 1.
Partial differential equations and Fourier series.
Introduction to Computer Science - C.

Selected Publications (* indicates equal contribution)

2026 Beyond Next Token Probabilities: Learnable, Fast Detection of Hallucinations and Data Contamination on LLM Output Distributions
Guy Bar-Shalom*, Fabrizio Frasca*, Derek Lim, Yoav Gelberg, Yftah Ziser, Ran El-Yaniv, Gal Chechik, Haggai Maron
(AAAI 2026)
A shorter version appeared @ Quantify Uncertainty and Hallucination in Foundation Models workshop @ ICLR 2025 (oral)

Neural Message-Passing on Attention Graphs for Hallucination Detection
Fabrizio Frasca*, **Guy Bar-Shalom***, Yftah Ziser, Haggai Maron
(Under review at ICLR 2026)

Selected Publications (* indicates equal contribution) (continued)

- A Graph Meta-Network for Learning on Kolmogorov–Arnold Networks
  Guy Bar-Shalom, Ami Tavory, Itay Evron, Maya Bechler-Speicher, Ido Guy, Haggai Maron
  (Under review at ICLR 2026)
- FS-KAN: Permutation Equivariant Kolmogorov-Arnold Networks via Function Sharing
  Ran Elbaz, **Guy Bar-Shalom**, Yam Eitan, Fabrizio Frasca, Haggai Maron
  (Under review at ICLR 2026)
- On The Expressive Power of GNN Derivatives
  Yam Eitan, Moshe Eliasof, Yoav Gelberg, Fabrizio Frasca, **Guy Bar-Shalom**, Haggai Maron
  (Under review at ICLR 2026)

2025

- Beyond Token Probes: Hallucination Detection via Activation Tensors with ACT-ViT
  Guy Bar-Shalom*, Fabrizio Frasca*, Yaniv Galron, Yftah Ziser, Haggai Maron
  (NeurIPS 2025)
- Balancing Efficiency and Expressiveness: Subgraph GNNs with Walk-Based Centrality
  Joshua Southern, Yam Eitan, **Guy Bar-Shalom**, Michael M. Bronstein, Haggai Maron, Fabrizio Frasca
  (ICML 2025)
- Topological Blindspots: Understanding and Extending Topological Deep Learning Through the Lens of Expressivity
  Yam Eitan, Yoav Gelberg, **Guy Bar-Shalom**, Fabrizio Frasca, Michael M. Bronstein, Haggai Maron
  Oral presentation
  (ICLR 2025)

2024

- A Flexible, Equivariant Framework for Subgraph GNNs via Graph Products and Graph Coarsening
  Guy Bar-Shalom*, Yam Eitan*, Fabrizio Frasca, Haggai Maron
  (NeurIPS 2024)
  A shorter version appeared @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024 (oral) winning the best paper award
- Subgraphormer: Unifying Subgraph GNNs and Graph Transformers via Graph Products
  Guy Bar-Shalom, Beatrice Bevilacqua, Haggai Maron
  (ICML 2024)
- Weakly-supervised Representation Learning for Video Alignment and Analysis
  Guy Bar-Shalom, George Leifman, Michael Elad
  (WACV 2024)

2023

- Window-Based Distribution Shift Detection for Deep Neural Networks
  Guy Bar-Shalom, Yonatan Geifman, Ran El-Yaniv
  (NeurIPS 2023)

2022

- TransBoost: Improving the Best ImageNet Performance using Deep Transduction
  Omer Belhasin*, **Guy Bar-Shalom***, Ran El-Yaniv
  (NeurIPS 2022)

Awards

2025

- Jacobs Qualcomm Excellence Scholarship
- Faculty Excellence Scholarship, Technion

2024

- best paper award @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024

Awards (continued)

- VATAT Prize for Academic Excellence, Technion

2018 – present
- Teaching Assistant Excellence award (4 times).

Professional service

2024
- Co-organizer of iGDL: Israeli Geometric Deep Learning workshop

2022-present
- Reviewer at NeurIPS, ICML, ICLR

Teaching experience

2018 – present
- Teaching Assistant at the Technion (department of Mathematics)
  Course: *Partial Differential Equations and Fourier Series*
  Average teaching score: 4.5/5.

2021 – present
- Teaching Assistant at the Technion (department of Computer Science)
  Course: *Algorithms 1*
  Average teaching score: 4.6/5.

2021
- Teaching Assistant at the Technion (department of Computer Science)
  Course: *Introduction to Computer Science*
  Average teaching score: 4.4/5.

Other

2012
- Played in the NCAA basketball league (USA), at Mercyhurst University (Erie, PA)

2012 – present
- Semi-pro basketball player in Israel

2013-2016
- Israel Defense Forces (IDF), Staff Sergeant